

Cybersecurity: Protecting the Digital World

Cybersecurity is the practice of protecting systems, networks, and programs from digital attacks. These cyberattacks typically aim to access, change, or destroy sensitive information, extort money, or disrupt normal business processes.

1. Common Threat Vectors

Phishing attacks trick users into revealing credentials or installing malware via deceptive emails or websites. Ransomware encrypts victim data and demands payment for decryption keys. SQL injection exploits database vulnerabilities. Zero-day vulnerabilities are flaws unknown to the vendor and exploited before patches exist.

2. Encryption and PKI

Encryption converts readable data into ciphertext. Symmetric encryption (AES) uses one key; asymmetric encryption (RSA, ECC) uses a public/private key pair. TLS/SSL secures data in transit on the web. Public Key Infrastructure (PKI) manages digital certificates that bind public keys to identities.

3. Zero Trust Architecture

Zero Trust is a security model that assumes no user or device is inherently trusted, inside or outside the network perimeter. It requires continuous verification, least-privilege access, and micro-segmentation. Organisations adopting Zero Trust saw a significant reduction in breach costs according to IBM's 2023 Cost of a Data Breach report.

4. Incident Response

A structured incident response plan includes: preparation, identification, containment, eradication, recovery, and lessons learned (PICERL). Security Operations Centers (SOCs) use SIEM tools to aggregate logs and detect anomalies in real time.

5. Regulatory Landscape

Regulations such as GDPR (EU), CCPA (California), HIPAA (healthcare, US), and PCI-DSS (payment cards) impose data protection obligations on organisations. Non-compliance can result in substantial fines. The SEC now requires public companies to disclose material cybersecurity incidents within four business days.